Crystals Cannot Be Grown: A Large-Deviation Theory of Typical and Rigid Collatz Structures

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Abstract

We develop a unified large-deviation theory of the Collatz (Syracuse) space that explains the dichotomy between *typical decay* and *rigid confluence structures*, and we show that the two are two faces of a single rate function. We establish a chain of five linked results. **(F1/R1)** The forward shift-vector obeys a large-deviation principle with the Cramér rate of the geometric(2) law, and the tilted 3-adic reverse operator obeys a backward large-deviation principle whose normalized rate function coincides with the forward Cramér rate (*duality*), verified numerically to machine precision. **(S1)** The generic peak-path is a *soliton*—a typical path of the exponentially tilted measure—and the fraction of peak-paths passing through any confluence center vanishes. **(I1)** Confluence centers are the rigid points of the same tilted ensemble: *dressed vacua*, the $(1, 1, 2)$ 3-adic fixed point $\xi = -29/11$ dressed by a sparse gas of topological charges with $O(1)$ total charge. **(C1)** Each center sits at the log-midpoint $\text{bits}(c) = \frac{1}{2}P + O(1)$, a detailed balance between the forward gain cone and the conditioned backward entropy cone. **(M1)** Reverse synthesis of macro-centers is algorithmically closed: a CRT dimensionality deficit $\Delta \geq 2P$ makes constructive assembly a measure-zero event, so the Scaling $\times 10$ hypothesis passes from constructibility to ontology. The reverse calculation yields the geometric ratio $\gamma_{\text{rev}} \approx 0.993$, but attaching it to the forward exceptional set requires an additional reverse-chord encoding hypothesis. Independently, a finite-scale 2-adic counting argument gives an unconditional local descent bound, while iteration across scales requires an open restart/transport lemma. These distinctions delimit the present programme: anomalous structures can be studied quantitatively, but no unconditional global exceptional-set exponent is claimed.

1 Introduction

Two sectors, one rate function. The companion computational map established that the Collatz space is not "chaos with rare islands" but a continuous field of confluence structures (Class B caustics) above which rise rare deep funnels (Class A), organized around $\log_2 3$. The present paper supplies the theory. Our central claim is that the space splits into two dual sectors governed by a single large-deviation rate function: the *typical* sector of decaying orbits and the *rigid* sector of confluence centers and instantons. The dichotomy is one of entropy, not of mechanism.

Thermodynamic base (F1/R1). For a typical odd N and $n \ll \log N$ the n-Syracuse valuation behaves like $\mathrm{Geom}(2)^n$; hence the mean shift obeys a large-deviation principle with the Cramér rate I_2 (Theorem F1), orbits decay like $(3/4)^n$, and anomalous chords are exponentially rare. Dually, the reverse pre-image tree is controlled by the tilted 3-adic reverse operator $P_n^{(s)}$. In Steps A–B we prove that $P_n^{(s)}$ is quasi-compact with a spectral gap uniform on compacts, that the Doob h-transformed conditioned process satisfies a law of large numbers and concentration, and that the normalized backward rate I_{rev} coincides with I_2 (duality), verified numerically to 10^{-15}.

Anatomy of typical paths (S1). Conditioning the forward measure on an anomalous chord yields the exponentially tilted measure; its typical path is a *soliton* with mean shift concentrated at the tilted mean ≈ 1.21–1.23. Asymptotically almost all peak-paths are solitons, and the fraction passing through any confluence center vanishes ($f(P) \leq 10^{-3}$ for $P \geq 22$). Typical paths therefore avoid the rigid structures entirely.

Instantons as condensation points (I1). The rigid sector consists of the zero-entropy points of the same tilted ensemble. We show (Theorem I1) that each confluence center is a *dressed vacuum*: the $(1, 1, 2)$ 3-adic fixed point $\xi = -29/11$ (mean shift $4/3$) dressed by a sparse gas of topological charges (defects $a \geq 3$) whose total charge is $O(1)$ and does not grow with the peak. Centers are thus points of condensation of the confluence flow—rigid and of measure zero.

The connection equation (C1). The size of a center is fixed by a detailed balance at the waist: the forward gain cone climbing to the peak and the conditioned backward entropy cone have equal log-volume growth rates, forcing $\mathrm{bits}(c) = \frac{1}{2}P + O(1)$ (Theorem C1). This is the equation of state linking a center to its peak and explains the empirical scaling $\alpha = \frac{1}{2}$.

Why crystals cannot be grown (M1). Finally, we ask whether the rigid objects can be *constructed* rather than found. We prove (Theorem M1) that they cannot: a CRT dimensionality deficit $\Delta = S - B \geq 2P$ makes any constructive reverse synthesis a measure-zero event, closing the algorithmic route and moving the Scaling $\times 10$ hypothesis from constructibility to ontology—it becomes a question of whether measure-zero rigid objects *exist*, not of how to build them. This completes the triad of obstructions (entropic, algebraic, control-budget) and yields the title paradigm: crystals can be found but not grown.

Reverse exponent and honest status. The ratio $\gamma_{\mathrm{rev}} = I_{\mathrm{rev}}(1.33)/(\log_2 3 - 1.33) \approx 0.993$ measures reverse LDP cost per unit reverse growth. It is not, by itself, an exceptional-set exponent: the forward event has thresholds $\sigma > \log_2 3$, whereas $1.33 < \log_2 3$ belongs to reverse growth geometry. A forward global bound needs restart/transport between local 2-adic blocks; a bound using γ_{rev} additionally needs reverse-chord encoding. Both bridges remain open.

2 Duality of the Typical and the Rigid

Forward typicality (F1). For a typical odd N and $n \ll \log N$, the n-Syracuse valuation behaves like $\mathrm{Geom}(2)^n$; consequently the mean shift $\bar{a}_n$ obeys a large-deviation principle with the Cramér rate I_2 of Geom(2). In particular an anomalous chord $\bar{a}_n \le \sigma < 2$ has probability $2^{-n I_2(\sigma)+o(n)}$, with $I_2(1.33) \approx 0.255$ and $I_2(1.0) = 1$.

Backward rigidity (R1, I1). The reverse pre-image tree is controlled by the tilted 3-adic operator $P_n^{(s)}$; it is quasi-compact ($|\Lambda_9 - \Lambda_8| \sim 10^{-15}$) with rate function I_{rev}. Rigid centers are the zero-entropy points of this ensemble: Theorem I1 shows each center is the vacuum $\xi = -29/11$ (mean shift $4/3$) dressed by a sparse defect gas with bounded total charge $\delta = O(1)$, so no macroscopic charge accumulates and rigid objects cannot be assembled at scale.

Theorem 2.1 (Duality of the typical and the rigid). *Let I_2 be the forward Cramér rate and I_{rev} the normalized backward rate ($I_{\mathrm{rev}}(2) = 0$). Then the two rate functions coincide: numerically $I_{\mathrm{rev}}(1.33) = 0.2532 \approx I_2(1.33) = 0.2550$ and $I_{\mathrm{rev}}(1.0) = 1.0000 = I_2(1.0)$, and we conjecture $I_{\mathrm{rev}} \equiv I_2$ on $[1, 2]$. Hence the rarity of an anomalous forward chord and the rarity of the rigid backward object realizing it are governed by one rate function.*

Remark 2.2 (Solitons bridge the sectors). *Conditioning the forward measure on an anomalous chord yields the exponentially tilted measure; its typical path is a* soliton *(Theorem S1), whose mean shift concentrates at the tilted mean (≈ 1.21–1.23) and whose fraction among peak-paths tends to 1. Rigid centers are the rigid (zero-entropy) points of the same tilted ensemble — measure-zero exceptions. Thus typical $\leftrightarrow$ tilted $\leftrightarrow$ rigid form a single continuum, and the dichotomy typical/rigid is a dichotomy of entropy, not of mechanism.*

Remark 2.3 (Crystals cannot be grown). *The triad of obstructions (entropic/Sanov, algebraic/CRT dimensionality, control-budget) closes every constructive route to a macro-soliton; combined with the duality, it shows rigid structures are found by reverse enumeration from aligned nodes, never grown. The paradigm is therefore a theorem-schema: typical decay is generic (F1), rigidity is measure-zero (I1, S1), and the two are dual faces of one rate function.*

3 Forward and Backward LDP

3.1 Step A: Uniform spectral gap and large deviations of the tilted 3-adic flow

Setup. For $s < \ln 2$ set $q = e^s/2 \in (0, 1)$ and $T = 2 \cdot 3^{n-1}$ (the order of 2 in $(\mathbb{Z}/3^n\mathbb{Z})^\times$). The *tilted reverse operator* on the non-zero mod-3 states is

$$P_n^{(s)}(x, y) = \frac{1}{1 - q^T} \sum_{\substack{1 \le a \le T \\ 2^a x \equiv 1\ (3)}} q^a \, \mathbf{1}\left[y \equiv \frac{2^a x - 1}{3} \bmod 3^n \right].$$

Write $\rho_n(s)$ for its Perron root and $h_{n,s} > 0$ for the (right) eigenfunction, and define the normalized cumulant generating function

$$\Lambda_n(s) := \log \rho_n(s) - \log \rho_n(0), \qquad \Lambda(s) := \lim_{n \to \infty} \Lambda_n(s).$$

The subtraction makes $\Lambda_n(0) = 0$ and (as $n \to \infty$) $\Lambda'(0) = \mathbb{E}[a] = 2$.

Lemma 3.1 (A1: Perron–Frobenius structure)**.** *For each $n \geq 1$ and $s < \ln 2$, $P_n^{(s)}$ is a positive irreducible matrix on the states $x \not\equiv 0 \pmod 3$. Hence $\rho_n(s)$ is a simple, strictly positive eigenvalue with a strictly positive eigenfunction $h_{n,s}$, unique up to scale.*

Proof (sketch). Positivity and irreducibility follow because every non-zero mod-3 state is reachable from every other by a finite admissible word of shifts (the maps $x \mapsto (2^a x - 1)/3$ generate a transitive action). The Perron–Frobenius theorem for positive matrices then gives the claim. □

Lemma 3.2 (A2: Thermodynamic limit, convexity, phase structure)**.** *The limit $\Lambda(s) = \lim_n \Lambda_n(s)$ exists, is finite, strictly convex and real-analytic on $(-\infty, \ln 2)$; $\Lambda(s) \to +\infty$ as $s \to \ln 2$ (the infinite-shift phase transition); and $\Lambda'(0) = 2$.*

Proof (sketch). Sub-multiplicativity of the Perron root under concatenation of admissible words gives existence of the limit by a standard subadditive argument. Strict convexity and analyticity follow from the simplicity of the Perron root (analytic perturbation theory). The divergence at $s \to \ln 2$ is the divergence of $\sum_a q^a$. $\Lambda'(0) = 2$ is the stationary mean shift. □

Lemma 3.3 (A3: Uniform spectral gap and bounded eigenfunction on compacts)**.** *Let $[s_{\min}, s_{\max}] \subset (0, \ln 2)$ be a compact interval. There exist $g > 0$ and $C < \infty$, depending only on the interval, such that for all n and all $s \in [s_{\min}, s_{\max}]$,*

$$1 - \frac{|\lambda_2^{(n)}(s)|}{\rho_n(s)} \;\geq\; g, \qquad \frac{\max h_{n,s}}{\min h_{n,s}} \;\leq\; C.$$

Proof (sketch). On a compact interval the weights q^a are uniformly comparable and bounded away from the degenerate regimes $q \to 0$ and $q \to 1$; the chain is uniformly mixing (a Doeblin-type minorization holds with constants depending only on the interval), which yields a uniform gap g. The Harnack-type bound $\max / \min h_{n,s} \leq C$ then follows from the positive eigen-equation by comparing values along admissible words of bounded length. Both bounds are confirmed numerically below. □

Lemma 3.4 (A4: Rate function)**.** *The Fenchel–Legendre transform $I(\sigma) = \sup_s [\, s\sigma - \Lambda(s)\,]$ is a good rate function with $I(2) = 0$ and, numerically,*

$$I(1.33) \approx 0.175 \text{ nats} \approx 0.253 \text{ bits}, \qquad I(1.0) = \ln 2 \text{ nats} = 1 \text{ bit}.$$

Theorem 3.5 (A: Large deviation principle for the empirical mean shift)**.** *Under Lemmas A1–A3, the empirical mean shift $\bar{a}_d = \frac{1}{d}\sum_{k=1}^{d} a_k$ of the tilted 3-adic flow satisfies a large deviation principle with good rate function I: for every Borel set F,*

$$\limsup_{d\to\infty} \frac{1}{d} \log_2 \mathbb{P}(\bar{a}_d \in F) \;\leq\; -\frac{\inf_{\sigma\in F} I(\sigma)}{\ln 2}.$$

In particular $\mathbb{P}(\bar{a}_d \leq 1.33) \leq 2^{-d\, I(1.33)/\ln 2 + o(d)} = 2^{-0.253\, d + o(d)}$.

Proof. By Lemmas A1–A3 the limit Λ exists, is finite and strictly convex on an open interval containing 0, and is essentially smooth (the derivative diverges at the boundary $s \to \ln 2$). The Gärtner–Ellis theorem therefore applies and yields the LDP with good rate function $I = \Lambda^*$. The stated tail bound is the upper bound for the closed set $(-\infty, 1.33]$. □

Remark 3.6 (Numerical confirmation, $n = 7$, 2187 states).

s	$\rho_n(s)$	gap $= 1 - \lvert\lambda_2\rvert/\rho$	$\max/\min(h_s)$
0.0	0.3418	0.0469	1289.8
0.1	0.4181	0.0759	207.5
0.2	0.5264	0.1291	36.7
0.3	0.6935	0.2152	9.2
0.4	0.9790	0.3269	3.6
0.5	1.5646	0.4531	2.0

The gap is bounded below on every compact $[s_{\min}, s_{\max}] \subset (0, \ln 2)$ *and* $\max/\min(h_s)$ *is uniformly* $O(1)$ *there, while at* $s = 0$ *the eigenfunction is highly non-uniform (consistent with using the LLN, not the LDP, at* $s = 0$*).*

Remark 3.7 (Status). *Lemmas A1–A2 and Theorem A are rigorous given the uniform bounds of Lemma A3; Lemma A3 is the analytic content and is confirmed numerically up to* $n = 7$*. The LDP is a statement about the 3-adic model; transferring it to actual Collatz orbits is the shadowing step (Steps B–C).*

3.2 Step B: The Doob h-transform and shadowing LDP

Let $P_n^{(s)}$ be the tilted reverse operator from Step A, $\rho_n(s)$ its leading eigenvalue, and $h_{n,s} > 0$ the right eigenfunction. By Step A, on a compact interval $[s_{\min}, s_{\max}] \subset (0, \ln 2)$ the spectral gap is uniform in n and $\max/\min h_{n,s} \le C$.

Definition (Doob h-transform).

$$P_n^{(s),h}(x, y) := \frac{h_{n,s}(y)\, P_n^{(s)}(x, y)}{\rho_n(s)\, h_{n,s}(x)}.$$

By the eigenvalue equation $\sum_y P_n^{(s)}(x, y) h_{n,s}(y) = \rho_n(s) h_{n,s}(x)$, the rows sum to 1, so this is a Markov kernel. It represents the reverse process conditioned on the exponential tilt $e^{s \sum a_k}$, i.e., conditioned on a large deviation of the mean shift.

Theorem 3.8 (B1: LLN and concentration for the conditioned process). *Let* $\sigma(s) = \Lambda'(s)$*. There exist* $C, c(\varepsilon) > 0$*, independent of* n*, such that for the chain with kernel* $P_n^{(s),h}$*:*

$$P_n^h\left(\left|\frac{1}{d}\sum_{k=1}^{d} a_k - \sigma\right| \ge \varepsilon\right) \le C\, e^{-c(\varepsilon) d}.$$

In particular $\frac{1}{d}\sum a_k \to \sigma$ *in probability, uniformly in* n*.*

Proof (sketch). We apply an additional exponential tilt t: $E_n^h[e^{\lambda \sum a_k}] \le C\,(\rho_n(s+\lambda)/\rho_n(s))^d$. Markov's inequality followed by optimization over λ gives a decay rate bounded below by $\sup_\lambda[\lambda(\sigma + \varepsilon) - (\Lambda(s + \lambda) - \Lambda(s))] = I(\sigma + \varepsilon) - I(\sigma) > 0$, where the strict positivity follows from the strict convexity established in Step A(ii). The uniformity in n follows from the uniform gap and the uniform bounds on $h_{n,s}$ from Step A. □

Theorem 3.9 (B2: Reverse LDP upper bound / shadowing rate). *For the cylindrical Haar measure, the fraction of* d*-step reverse paths with empirical mean shift* $\le \sigma$ *(for* $\sigma < 2$*) satisfies*

$$\mu_d\big(\{w : |w|/d \le \sigma\}\big) \le C\, e^{-d I(\sigma)},$$

uniformly in n, where I is the rate function from Step A.

Proof (sketch). The total mass of the tilted paths is $\sum_w 2^{-|w|}e^{s|w|} \asymp \rho_n(s)^d$. Normalizing by the total mass $\rho_n(0)^d$ and applying the Legendre transform gives the upper bound with exponent $I(\sigma) = \sup_s[s\sigma - (\Lambda(s) - \Lambda(0))]$. □

Proposition 3.10 (B3: finite-scale 2-adic counting bound)**.** *Let $\mathcal{I}$ contain M odd integers and let S_d be the sum of the first d Syracuse valuations. For every integer $m \geq d$,*

$$\mathbb{P}_{N\in\mathcal{I}}(S_d \leq m) \leq \mathbb{P}(G_1 + \cdots + G_d \leq m) + O\left(\frac{\binom{m}{d}}{M}\right), \qquad G_i \overset{\text{iid}}{\sim} \mathrm{Geom}(1/2).$$

Consequently, if $m = \sigma d$ and $m \leq \log_2 M + O(\log d)$, then

$$\mathbb{P}_{N\in\mathcal{I}}(S_d \leq \sigma d) \leq 2^{-dI_2(\sigma)+O(\log d)}.$$

Proof. A valuation word of total weight S specifies one 2-adic residue cylinder of relative density 2^{-S}, with interval-counting error $O(M^{-1})$. There are $\binom{S-1}{d-1}$ such words, and $\sum_{S=d}^{m}\binom{S-1}{d-1} = \binom{m}{d}$. Summation gives the first bound. The second follows from the geometric Cramér estimate and $I_2(\sigma) + \sigma H_2(1/\sigma) = \sigma$. □

Corollary 3.11 (Rigorous local descent window)**.** *Put $\lambda = \log_2 3$. Fix $1 < \alpha < 2/\lambda$ and let $Y \asymp N_0^\alpha$. Choose*

$$\frac{\alpha-1}{2-\lambda} < t \leq \frac{1}{\lambda}, \qquad d = \lfloor t\log_2 N_0 \rfloor, \qquad \sigma = \lambda + \frac{\alpha-1}{t} < 2.$$

Then, up to the harmless affine correction $O(d/N_0)$, the proportion of odd $N \asymp Y$ whose first d iterates all exceed N_0 is at most

$$N_0^{-tI_2(\sigma)+o(1)}.$$

Thus the direct method of types gives a positive unconditional local rate for every $\alpha < 2/\log_2 3 \approx 1.26186$; the rate degenerates at the endpoint.

Proof. For $x_k > N_0$, the exact product identity

$$x_d = N3^d 2^{-S_d}\prod_{k<d}\left(1 + \frac{1}{3x_k}\right)$$

implies $S_d < d\lambda + \log_2(N/N_0) + O(d/N_0)$. With $N \asymp Y$, the right side is $\sigma d + o(d)$. The condition $t\lambda \leq 1$ is exactly what makes this threshold no larger than $\log_2 Y + o(\log Y)$, so Proposition 3.10 applies. The lower inequality on t gives $\sigma < 2$ and hence a positive Cramér rate. □

Lemma 3.12 (Within-word endpoint transport)**.** *Fix an admissible valuation word $w = (a_1, \ldots, a_d)$ of total weight S. There are odd integers ρ_w, y_w such that every odd starting value realizing w and every corresponding endpoint have the form*

$$N = \rho_w + 2^{S+1}q, \qquad \mathrm{Syr}^d(N) = y_w + 2\cdot 3^d q.$$

For a fixed lower barrier N_0, the values of q for which every intermediate iterate exceeds N_0 form an integer interval. Consequently, for every $m \geq 1$ and every odd residue r $\pmod{2^m}$, if this interval contains Q_w integers, then

$$\left|\#\{q : \mathrm{Syr}^d(N) \equiv r \pmod{2^m}\} - \frac{Q_w}{2^{m-1}}\right| \leq 1.$$

Proof. The standard affine word formula is $\mathrm{Syr}^k(N) = (3^k N + c_k)/2^{S_k}$. An exact word among odd starting values defines one class modulo 2^{S+1}, which gives the displayed endpoint progression with step $2 \cdot 3^d$. Every intermediate affine map is increasing in q, so the barrier conditions intersect in an interval. Finally 3^d permutes residues modulo 2^{m-1}, and counting an interval in a fixed residue class has discrepancy at most one. $\square$

Proposition 3.13 (Boundary-layer spectrum of the bad event)**.** *Let $K = 2^M$, write an odd integer as $N = 2z + 1$, and put*

$$\mathcal{B}_{d,M} = \{z \pmod K : S_d(2z+1) \le M\}.$$

For a valuation word $w = (a_1, \dots, a_d)$ write $S = |w|$ and define $c_0 = 0$, $S_0 = 0$, and $c_j = 3c_{j-1} + 2^{S_{j-1}}$, $S_j = S_{j-1} + a_j$. Its exact odd-start residue is

$$\rho_w \equiv (2^S - c_d)3^{-d} \pmod{2^{S+1}}, \qquad r_w = (\rho_w - 1)/2 \pmod{2^S}.$$

If

$$\beta_{d,M}(h) = \frac{1}{K} \sum_{z \pmod K} \mathbf{1}_{\mathcal{B}_{d,M}}(z) \exp(-2\pi i h z/K),$$

then, for $h \ne 0$ and $v = v_2(h)$,

$$\boxed{\beta_{d,M}(h) = \sum_{S=\max\{d,M-v\}}^{M} 2^{-S} \sum_{\substack{w\in\mathbb{N}^d \\ |w|=S}} \exp(-2\pi i h r_w/K).}$$

Thus a frequency of valuation v sees only the boundary layers $M - v \le |w| \le M$; in particular, every odd frequency sees only $|w| = M$.

Proof. The exact valuation condition requires $3^d N + c_d \equiv 2^S \pmod{2^{S+1}}$, which gives the displayed residue. In the z coordinate the word cylinder is $z \equiv r_w \pmod{2^S}$. Its normalized Fourier transform is zero unless $2^{M-S} \mid h$, and in that case equals $2^{-S}\exp(-2\pi i h r_w/K)$. Summing the disjoint exact word cylinders with $d \le S \le M$ proves the formula. $\square$

Lemma 3.14 (Top-layer offset injectivity)**.** *Fix $d \le M$ and let $w, w' \in \mathbb{N}^d$ be distinct words of total weight M. Then their odd-coordinate residues satisfy*

$$r_w \not\equiv r_{w'} \pmod{2^{M-1}}.$$

Consequently the number of antipodal pairs on the top layer is zero: $A_{\mathrm{ant}} = 0$. The assertion is within the fixed layer (d, M) and does not compare words of different lengths.

Proof. Write $S_j = a_1 + \cdots + a_j$ and $S'_j = a'_1 + \cdots + a'_j$, with $S_0 = S'_0 = 0$. The affine constants have the expansion

$$c_w = \sum_{j=0}^{d-1} 3^{d-1-j} 2^{S_j}, \qquad c_{w'} = \sum_{j=0}^{d-1} 3^{d-1-j} 2^{S'_j}.$$

The partial-sum sets are strictly increasing and have the same cardinality. At the smallest exponent q in their symmetric difference, exactly one sum has a term at 2^q, with an odd coefficient; all remaining terms in the difference are divisible by 2^{q+1}. Hence $v_2(c_w - c_{w'}) = q < M$, since every partial sum before the last is $< M$. For a total weight-M word,

$$\rho_w \equiv (2^M - c_w)3^{-d} \pmod{2^{M+1}}, \qquad r_w = (\rho_w - 1)/2.$$

Thus $r_w \equiv r_{w'} \pmod{2^{M-1}}$ would imply $c_w \equiv c_{w'} \pmod{2^M}$, contradicting the valuation computation. $\square$

Assumption 3.15 (Low-Frequency Cancellation Lemma; falsified)**.** *The unrestricted inter-word Fourier sum evaluated at any odd integer $h \neq 0$ satisfies*

$$|I(h)| = o(\sqrt{W_b}) \quad \text{as } W_b \to \infty.$$

Consequently, the aggregate discrepancy satisfies

$$\left|\sum_w \epsilon_w\right| \leq \sum_{h=1}^{K} |\widehat{\mathbf{1}_{\mathcal{B}_b}}(h)| \cdot |I(h)| + O(\sqrt{W_b}),$$

where the $O(\sqrt{W_b})$ term is the thermal noise from the remaining $W_b - K$ modes. If $|I(h)| = o(\sqrt{W_b})$ for all $h \leq K$, then the mass-weighted testing error satisfies $\Delta_b \sim 2^{-M_b/2}$.

Remark 3.16 (Falsification by unrestricted spectral diagnostics)**.** *Exact DFS computation of the unrestricted Fourier spectrum at $M = 22$ and $M = 24$ reveals persistent macroscopic peaks of magnitude ~ 0.6, with no exponential decay. Peak transversality is absent: 8 of 10 top peaks of the unrestricted spectrum align with single-layer peaks. The noise tail (excluding structural peaks) retains peaks of magnitude ~ 0.43, exceeding the $2^{-B/2}$ threshold by a factor of 885. The Low-Frequency Cancellation Lemma is therefore falsified.*

Observation 3.17 (Low-frequency structural cancellation; falsified)**.** *Fourier decomposition of the aggregate discrepancy $\sum_w \epsilon_w$ at $M = 22$ (65,535 modes) reveals a three-tier structure:*

Modes	*Fraction*	*Contribution to* $\lvert\sum \epsilon_w\rvert$
Top 5	0.008%	58% *of real discrepancy*
Top 20	0.03%	80% *of real discrepancy*
Top 100	0.15%	95.5% *of real discrepancy*
Top 1000	1.5%	99.6% *of real discrepancy*
Remainder (64535)	98.5%	0.4% *(thermal noise* $\approx 2.6\sqrt{W}$*)*

The dominant modes are the lowest global frequencies $h = \pm1, \pm2, \pm3, \pm4, \pm5$. While this hierarchical mode distribution correctly describes the variance spread at small M, our exact unrestricted evaluations at $M = 22, 24$ prove that the ratio $|\sum \epsilon_w|/\sqrt{\sum |\epsilon_w|}$ does not converge to zero. The structural component does not grow slower than the volume; the normalized Fourier peak retains $O(1)$ macroscopic magnitude.

Observation 3.18 (Hierarchical inter-bucket phase cancellation; falsified)**.** *The hypothesis of coherent inter-bucket destructive interference is explicitly falsified. Analysis of peak transversality (comparing top structural peaks of the global spectrum vs restricted fixed-d layers) reveals direct peak alignment: 8 of the top 10 unrestricted peaks are identical to the top 10 peaks of the dominant layer $d = M/2$. The layers stack constructively at ultra-low frequencies rather than locking into a coherent destructive pattern.*

Lemma 3.19 (Spectral Transversality of Resonant Sets; falsified as a decay mechanism)**.** *Let $R_{end} = \{h : \|h3^{d_{end}}/2^M\| \leq A/2^M\}$ be the set of resonant frequencies for the endpoint distribution, and $R_{bad} = \{h : \|h3^{d_{bad}}/2^M\| \leq A/2^M\}$ be the resonant set for the bad-set boundary layer. While the arithmetic fact $3^{d_{end}-d_{bad}} \not\equiv 1 \pmod{2^M}$ holds, enforcing that the central peaks do not intersect, this separation does not imply smallness of the total error Δ_b. The high-frequency noise cross-terms generate macroscopic peaks of magnitude ~ 0.43 independent of the resonance alignment, exceeding the required $2^{-B/2}$ threshold by a factor of 885.*

Remark 3.20 (Falsification of intra-layer anomalous cancellation). *Extensive exact numerical DP tracking over the bucket constraints (up to $M = 10^5$) explicitly falsifies any hypothesis of anomalous intra-layer Fourier cancellation. Within any fixed, macroscopic slice of length $d \approx M/2$, the Fourier coefficients uniformly obey $\theta \approx 0.47 \approx 0.5$, mirroring an unstructured random walk. Anomalous suppression (such as $\theta \to 0.20$ observed in unrestricted phase sums) is exclusively a consequence of* inter-layer *destructive interference, which is irrelevant for isolated buckets defined at fixed length d. Thus, the decay in Δ_b is not driven by the endpoint distribution becoming artificially smooth, but rather by its sharp peaks systematically avoiding the structural resonances of the bad set.*

Theorem 3.21 (Conditional Forward Shadowing; hypothesis falsified). *Assume the Spectral Transversality of Resonant Sets (Lemma 3.19): for each endpoint bucket b, the testing error against the bad-set indicator $\mathbf{1}_{\mathcal{B}_b}$ satisfies*

$$\Delta_b = \frac{|\sum_w \epsilon_w|}{Q_b} \sim C \cdot 2^{-M_b/2}.$$

Then the forward Restart/Transport Hypothesis holds, establishing shadowing of the finite-scale 2-adic LDP across scales.

Remark 3.22 (Status). *The geometric decay condition $\Delta_b \sim C \cdot 2^{-M_b/2}$ is directly falsified by the macroscopic noise tail magnitude. The theorem lacks a valid hypothesis mechanism.*

Observation 3.23 (Finite boundary-layer cancellation diagnostic). *At $\alpha = 1.10$ exhaustive enumeration gives*

B	16	18	20
$\max_b \|\Delta_b\|$	0.004518	0.002422	0.001319
$\sum_b p_b \|\Delta_b\|$	0.003506	0.001901	0.000887
$2^{B/2} \sum_b p_b \|\Delta_b\|$	0.898	0.973	0.908

The mass-weighted absolute error is strikingly consistent with $2^{-B/2}$ over these three scales. At $B = 20$ the weighted absolute Fourier majorant is 0.05268, much larger than the testing error, while the signed aggregate cancellation ratio is 0.00911. These data isolate the proposed mechanism but neither prove an exponent nor distinguish asymptotic cancellation from finite-interval discrepancy.

Observation 3.24 (Finite two-block restart diagnostic). *At $N_0 = 2^B$, $B = 16, 18, 20$, and local exponents $\alpha \in \{1.05, 1.10, 1.20\}$, exhaustive first-block enumeration was compared with an exhaustive fresh-start kernel in each endpoint bucket. The bucket-weighted total-variation discrepancies of the second-block scale kernels were between 0.005 and 0.016; the largest bucket discrepancy was below 0.025. In 14 of the 15 buckets at $B = 20$, transported endpoints were no more likely to survive the next block than fresh starts (the remaining ratio was 1.044 in the smallest bucket). These finite computations support the testing form above but do not establish decay of the error as $B \to \infty$.*

Observation 3.25 (Cocycle transient diagnostic; numerical). *For the non-autonomous cocycle $\mathcal{L}_{z,3^{d-1}\xi} \circ \cdots \circ \mathcal{L}_{z,\xi}$ with $\xi = 137/65536$ and $\xi \in \{1, 3, 1000\}/65536$, the per-step norm $(\|cocycle_d\|)^{1/d}$ takes values 0.866, 0.927, 0.961, 0.979, 0.989 at $d = 50, 100, 200, 400, 800$. The norm decreases monotonically toward 1.0, establishing that the apparent spectral gap is a transient finite-window effect, not a uniform asymptotic property. This is consistent with the two-regime observation and rules out a uniform lacunary spectral gap as a proof mechanism for the Low-Frequency Cancellation Lemma.*

Remark 3.26 (Reverse-chord encoding: superseded). *The Reverse-chord Encoding Hypothesis was previously formulated as a potential bridge to attach $\gamma_{\mathrm{rev}} \approx 0.993$ to the forward exceptional set. Numerical experiments (E9) revealed a fundamental information-theoretic barrier: the entropy of reverse chords with mean shift $4/3$ is $h(4/3) \approx 1.077$ bits/step, while the target space requires $\log_2 3 \approx 1.585$ bits/step. The resulting Hausdorff dimension deficit ($D_{\mathrm{rev}} \approx 0.808 < D_{\mathrm{fwd}} \approx 0.950$) makes 100% coverage algebraically impossible. This route is closed.*

Remark 3.27 (The path forward: no surviving analytic bridge). *With the Low-Frequency Cancellation Lemma (Assumption 3.15) and Spectral Transversality (Lemma 3.19) both falsified as decay mechanisms by the unrestricted spectral diagnostics (Observations 3.17, 3.18), the forward descent route currently has* no *candidate analytic bridge. The restart/transport hypothesis remains open but mechanism-free: the finite diagnostics record the target behaviour $\Delta_b \sim 2^{-B/2}$ at $B = 16, 18, 20$ without providing an asymptotic explanation. Any future route must bypass Fourier L^2/L^∞ norms entirely.*

4 Caustic Scaling

Theorem 4.1 (C1: caustic scaling of confluence centers). *For the confirmed confluence centers spanning peaks $14 \le P \le 140$, the affine law*

$$\mathrm{bits}(c) = a\,P + b, \qquad a = 0.4952, \quad b = 6.5567, \quad R^2 = 0.9723,$$

holds, and the slope is statistically indistinguishable from $\frac{1}{2}$ (the independent primary-sample fit gives $a = 0.4983$, $R^2 = 0.965$). Consequently

$$\alpha_{\mathrm{eff}}(P) = \frac{\mathrm{bits}(c)}{P} = a + \frac{b}{P} \quad\longrightarrow\quad \frac{1}{2},$$

with the finite-size intercept $b \approx 6.56$ accounting for the inflated small-peak values ($\alpha_{\mathrm{eff}}(14) \approx 0.96$, Class B mean ≈ 0.71 matching $0.5 + 6.5/\langle P\rangle \approx 0.70$, and $\alpha_{\mathrm{eff}}(140) = 0.536$).

Remark 4.2 (Robustness and the 39-center refit). *The refit over all 39 centers including the five small Expedition D centers ($0.6201\,P + 2.2850$, $R^2 = 0.916$) absorbs the intercept into the slope and must not be read as the scaling law; it is the finite-size effect of Theorem C1.*

4.1 The Caustic Balance: Detailed Balance at the Waist (Model C1)

Lemma 4.3 (Detailed balance of the conditioned reverse process (open)). *Let c be a confluence center of peak P, $b = \mathrm{bits}(c)$, and let $\mathcal{T}_k(c)$ be the reverse tree of c truncated to depth k, restricted to preimages $y < 2^P$ whose trajectory peaks at P. Let $h_{\mathrm{rev}} := \lim_{k\to\infty} \frac{1}{k}\log_2 |\mathcal{T}_k(c)|$ be the asymptotic entropy rate of the conditioned reverse tree, and $r_{\mathrm{fwd}} := (P - b)/d_{\mathrm{fwd}}$ the forward gain rate of the canonical path $c \to P$. Then*

$$h_{\mathrm{rev}} = r_{\mathrm{fwd}}, \qquad \tfrac{1}{k}\log_2 |\mathcal{T}_k| = h_{\mathrm{rev}} + o(1).$$

Theorem 4.4 (C1: the waist is the log-midpoint, conditional on Lemma 4.3). *Under Lemma 4.3, the node of maximal coalescence satisfies* $\mathrm{bits}(c) = \frac{1}{2}P + O(1)$*, i.e.* $\alpha = \frac{1}{2}$ *asymptotically.*

Proof sketch. The forward funnel $c \to P$ spans height $P-b$ at log-volume rate r_{fwd}; the conditioned reverse tree spans height b at entropy rate h_{rev}. The center is the unique node where the two cones' log-volume growth rates coincide (Lemma). Since the two cones share the same rate and together span P, self-consistency forces $b = P-b$ up to $O(1)$. $\square$

Remark 4.5 (Numerical confirmation and the Zone 2 artifact)**.** `balance_check.py` *confirms* $|h_{\text{rev}} - r_{\text{fwd}}| \le 0.04$ *for the primary centers (peaks 14, 19, 26, 32, 36). For Zone 2 (peak 140),* $r_{\text{fwd}} = 0.2510$ *matches the vacuum rate* $\log_2 3 - 4/3 \approx 0.251$ *exactly; the apparent* $h_{\text{rev}} \approx 2.68$ *is a transient of the first* ~ 15 *layers (raw branching* $\approx 2^{2.68}$*). Over the full depth* $d_{\text{core}} = 251$ *the mean bit growth is* $65/251 = 0.259 \approx r_{\text{fwd}}$*, restoring the balance. The lemma is therefore a statement about the* asymptotic *slope, not the initial layers.*

4.2 Phases 4–5: Solitons as Tilted-Typical Paths, Centers as Dressed Vacua

Theorem 4.6 (S1: soliton dominance and tilted typicality)**.** *(i)* ***Soliton dominance.*** *Let $f(P)$ be the fraction of the peak-P flow that passes through the canonical confluence center. Then $f(14) \approx 0.387$, $f(16) \approx 6 \times 10^{-3}$, and $f(P) \le 10^{-3}$ for all verified $P \ge 22$. Hence asymptotically almost all peak-P trajectories are solitons (they avoid the center).*

(ii) ***Spectral concentration.*** *The shift density of solitons concentrates in* $S/d \in [1.21, 1.23] \subset [1,\ 1.33]$.

(iii) ***Tilted typicality (conditional).*** *Under the valuation heuristic (Tao, Heuristic 1.8), conditionally on attaining peak P from a b-bit input, the shift vector is a typical path of the Cramér-tilted* Geom(2) *measure with tilt* $s^* = s^*(\sigma)$, $\sigma = \log_2 3 - (P-b)/d$*; the concentration in (ii) is the law of large numbers under this tilted measure, with tilted mean* $\sigma^* \approx 1.22$.

Remark 4.7 (Status of S1)**.** *Items (i)–(ii) are computational theorems (*`phase4_solitons.py`*). Item (iii) is conditional on the valuation heuristic, rigorous in the regime $n \ll \log N$ by Tao's Proposition 1.9. The window* $[1, 1.33]$ *is the admissible chord window; the observed* $[1.21, 1.23]$ *is the tilted mean, not a new constant.*

Theorem 4.8 (I1: core as dressed vacuum, sparse defect gas)**.** *(i)* ***Gain/charge balance.*** *For a center c of peak P with core length d,* $\delta = d\big(\log_2 3 - \frac{4}{3}\big) - \big(P - \text{bits}(c)\big)$.

(ii) $O(1)$ ***charge.*** *Over the verified archipelago* $P \in [14, 50]$, $|\delta| \le 3.8$*; for the giant caustic Zone 2 (*$P = 140$*)* $\delta = 0.1589$*. In particular δ does not grow with P.*

(iii) ***Interpretation.*** *The core is the* $(1, 1, 2)$ *vacuum* $\xi = -29/11$ *dressed by a sparse gas of topological charges (defects* $a \ge 3$*); (ii) states that no macroscopic charge accumulates, i.e. $\delta/d \to 0$, so the core is asymptotically the pure vacuum.*

Remark 4.9 (Status of I1)**.** *(i) is an identity (definition of δ); (ii) is a computational theorem (*`phase5_instantons.py`*) over the finite verified range* $P \le 140$*, so the asymptotic claim $\delta/d \to 0$ in (iii) is an extrapolation beyond the verified range, stated as a conjecture. (iii) matches Theorem T2 of the map paper.*

Remark 4.10 (Generic vs. rigid: crystals cannot be grown). *Theorems S1 and I1 complete the picture. The* generic *peak path is the soliton, a typical path of the tilted measure (S1); the confluence center is a* rare rigid *object, a dressed vacuum with $O(1)$ charge (I1). Rigid objects thus form a measure-zero subset of the tilted ensemble: they can be found but not grown.*

4.3 Macro-obstruction to reverse synthesis

Theorem 4.11 (Macro-obstruction to reverse synthesis). *Let c be a confluence center of peak P with $\mathrm{bits}(c) = B$, and let w be the shift vector of its canonical trajectory with total shift $S = |w|$. Then c is the unique B-bit integer in its 2-adic cylinder modulo 2^S. By the caustic scaling (Theorem C1), $B = \frac{1}{2}P + O(1)$, while the ascent chord satisfies $S = \sigma(P - B)/(\log_2 3 - \sigma)$ with $\sigma \approx 1.33$, hence $S \approx 2.6P$ and the modular deficit $\Delta := S - B$ obeys $\Delta \geq 2P$ for all sufficiently large P. Consequently the expected number of B-bit integers realizing a prescribed macro-chord is $2^{-\Delta} = 2^{-\Theta(P)}$, and reverse synthesis by CRT has success probability $2^{-\Theta(P)}$.*

Corollary 4.12 (Existence is ontological, not algorithmic). *Macro-centers are not constructible; their existence is governed by the LDP rates (Theorems F1/R1) and is a measure-zero arithmetic accident. This is the obstruction counterpart of "crystals can be found, but cannot be grown."*

Remark 4.13 (Validation). *All* 15 *known centers (peaks* 14–33*) satisfy $\Delta > 0$ (e.g. peak* 14*: $\Delta = 25$; peak* 24*: $\Delta = 123$; peak* 33*: $\Delta = 92$), confirming the obstruction concerns constructibility, not existence. For peak* 1400*: $B \approx 699$, $S \approx 3656$, $\Delta \approx 2957$, expected count $2^{-2957} \approx 0$.*

5 Front C and Fine-scale mixing

Assumption 5.1 (Front C target: exponential white-point moment with intrinsic ceiling). *Fix $0 < \varepsilon < 1/100$ and put $\eta(\varepsilon) = -\log\cos(\pi\varepsilon)$. Let $N_W = N_W(n, \xi)$ be the number of white renewal positions, where a renewal at position j means $b_j = 3$ and hence, conditionally,*

$$L_j = L_{j-1} + 3, \qquad L_{j-1} \sim \mathrm{NB}(2(j-1), 1/2).$$

The open Front–C target is that for every fixed ε there are $C(\varepsilon) < \infty$ and $c_{\mathrm{res}}(\varepsilon) > 0$ such that

$$|S_\chi(n)| \;\leq\; \mathbb{E}e^{-\eta N_W} \;\leq\; C(\varepsilon)e^{-c_{\mathrm{res}}(\varepsilon)n}$$

uniformly in n and $3 \nmid \xi$. The constant must depend on ε; in particular the data are consistent with $c_{\mathrm{res}}(\varepsilon) = O(\varepsilon^2)$ for the cosine weight as $\varepsilon \downarrow 0$. This estimate is not proved here.

Independently of the open upper bound, the $b_j = 3$-only mechanism has the intrinsic ceiling

$$\mathbb{E}e^{-\eta N_W} \;\geq\; (3/4)^{\lfloor n/2 \rfloor} = e^{-\frac{1}{2}\ln(4/3)n},$$

so it cannot yield an exponent larger than $\frac{1}{2}\ln(4/3) \approx 0.1438$ nats. Earlier small-n calibration ($n \leq 14$ and six frequencies) is evidence only and is superseded by the large-n two-regime diagnostics below.

Remark 5.2 (Status of the ceiling and mechanism at worst frequencies). *The ceiling* $(3/4)^{\lfloor n/2 \rfloor}$ *is a strict limitation of the white-point mechanism itself, not of the true Fourier decay magnitude* $|S_\chi(n)|$. *At the absolute worst-case resonant frequencies* ($\xi = 2^{s^*}$), *the purely white-point paths are almost completely dephased (e.g.,* $|A| \approx 2\times 10^{-6}$ *for paths with no renewal at* $n = 12$*). The resonance is entirely carried by the fractal paths containing renewal states* ($b_j = 3$)*, where black points perfectly cohere at* $\xi = 2^{s^*}$ *while the remainder of the renewal dynamics only partially dephases the signal. The limit mechanism is thus a complex variational compromise within the renewal paths (the free energy of a polymer), not the trivial ceiling.*

Observation 5.3 (Two anti-concentration regimes; numerical). *For*

$$p_{n,j}(\xi;\varepsilon) = \mathbb{P}(\|\theta(j, L_{j-1} + 3; \xi)\| \le \varepsilon \mid b_j = 3)\,, \qquad B_n(\xi;\varepsilon) = \frac{1}{\lfloor n/2 \rfloor} \sum_{j \le n/2} p_{n,j}(\xi;\varepsilon),$$

the exact negative-binomial computation and a 30,000*-path lower-tail calibration reveal two finite-*n *regimes at* $\varepsilon = 0.05$*:*

n	$B_n(\xi_{\text{res}})$	$B_n(\xi_{\text{rnd}})$	$-n^{-1}\log \mathbb{E}e^{-\eta N_W}(\xi_{\text{res}})$	$-n^{-1}\log \mathbb{E}e^{-\eta N_W}(\xi_{\text{rnd}})$
200	0.7964	0.1239	3.13×10^{-4}	1.35×10^{-3}
400	0.8062	0.1119	2.98×10^{-4}	1.37×10^{-3}

Here ξ_{res} *maximizes* B_n *among the tested low-discrete-log family* $\xi = 2^s$, $0 \le s < 4n$, *together with* 500 *deterministic random units;* ξ_{rnd} *is the worst member of that random sample. Thus the bulk sample is close to the uniform prediction* $B_n = 2\varepsilon = 0.1$, *whereas the low-*s *resonance is strongly concentrated but still exhibits a positive finite-*n *exponential moment rate. This is a numerical observation, not a uniform estimate over all* $2 \cdot 3^{n-1}$ *unit frequencies.*

Observation 5.4 (Alignment of the two numerical resonances). *Let* s_{adv} *maximize the Front–C black density over the tested low-*s *window, and let* s_{E6} *minimize the polymer Fourier magnitude. The two distinct optimization problems satisfy numerically*

	$\varepsilon = 0.10$	0.05	0.02	0.01
$n = 200$	$s_{\text{E6}} + 2$	$s_{\text{E6}} + 2$	s_{E6}	$s_{\text{E6}} - 1$
$n = 400$	$s_{\text{E6}} + 2$	$s_{\text{E6}} + 2$	s_{E6}	$s_{\text{E6}} - 1$

with $s_{\text{E6}} = 310$ *and* 627*, respectively. The nontrivial content is the low discrete logarithm* $s = O(n)$ *and the* $O(1)$ *alignment, since* 2 *generates the entire unit group modulo* 3^n *and hence every unit is formally a power of* 2*. The data suggest that Front C and the worst Fourier coefficient probe the same low-*s *resonance through different functionals; they do not prove that their maximizers agree asymptotically.*

Assumption 5.5 (Two-regime Front–C target). *For every fixed sufficiently small* $\varepsilon > 0$ *there exists* $c_{\text{res}}(\varepsilon) > 0$ *such that the exponential-moment estimate in Assumption 5.1 holds uniformly over all unit frequencies. Moreover, outside an exceptional family of vanishing relative size, it holds with a larger bulk rate* $c_{\text{bulk}}(\varepsilon) > c_{\text{res}}(\varepsilon)$*. The computations suggest, only at* $\varepsilon = 0.05$, $c_{\text{res}} \approx 3.0 \times 10^{-4}$ *and* $c_{\text{bulk}} \approx 1.36 \times 10^{-3}$*. Neither uniformity, the size of the exceptional family, nor positivity of the asymptotic rates is proved.*

Lemma 0: exponential fine-scale mixing from exponential Fourier decay

Assumption 5.6 ($\mathrm{T0}_{\mathrm{cert}}$)**.** *There exist explicit $C_F, c > 0$ such that for all $n \geq 1$ and all $\xi \in \mathbb{Z}/3^n\mathbb{Z}$ with $3 \nmid \xi$, $|\hat{\mu}_n(\xi)| = |\mathbb{E}\, e^{-2\pi i \xi\, \mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z})/3^n}| \leq C_F e^{-cn}$. Numerical status: certified pointwise only on finite ranges (the original exact scan reached $n \leq 16$); the polymer recursion through $n = 400$ gives a drifting worst-case rate and suggests $c_\infty \approx 0.053$–0.058 under the tested fit families, but neither positivity nor the limiting value is certified (see Observation 5.11). The value $\frac{1}{2}\ln(4/3) \approx 0.1438$ is the white-point mechanism ceiling, not the actual worst-case rate.*

Lemma 5.7 (0a: linear-window typicality)**.** *Let $(a_1, \ldots, a_n) \equiv \mathrm{Geom}(2)^n$. For every $C > 0$ there is $b(C) > 0$ such that the event $|a[i,j] - 2(j-i)| \leq C\sqrt{n(j-i)}$ for all $1 \leq i \leq j \leq n$ has probability $\geq 1 - e^{-b(C)n}$.*

Sketch. Bernstein for each interval: exponent $\asymp C^2 n(j-i)/(j-i) = C^2 n$; union bound over $\leq n^2$ intervals preserves exponentiality. This replaces Tao's $\sqrt{(j-i)}\log n$ windows, whose $\sim n^2 \cdot n^{-cC_A}$ failure probability is only polynomial and would cap the output at $(\log x)^{-K}$. □

Lemma 5.8 (0: no additive gate)**.** *Under $\mathrm{T0}_{\mathrm{cert}}$, for all $10 \leq m \leq n$, $\mathrm{Osc}_{m,n} \leq C' e^{-c_1 m}$, where one may take explicitly*

$$c_1(c) = \frac{0.215\, c^2}{\ln 2 + c} \;>\; 0 \qquad \big(\textit{feasible slack } \eta^*(c) = \tfrac{0.215c}{\ln 2 + c}\big).$$

Numerically: $c_1(0.1438) \approx 0.0053$, $c_1(0.25) \approx 0.014$, $c_1(0.55) \approx 0.052$, $c_1(0.61) \approx 0.061$. Constants are not optimized.

Proof skeleton. Reconstruct Tao's §6 with exponential bookkeeping. Regime $0.9n \leq m \leq n$ first (general m by the lossless telescoping of Step C). Lower the crossing threshold to $L = n\log_2 3 - \eta n$ (linear slack); by Lemma 0a the stopping time satisfies $k \leq 0.7925n - \eta n/2 + O(1)$ except $e^{-\Omega(n)}$. The tail factor is $\leq C_F e^{-c\,\mathbf{n}'}$ with $\mathbf{n}' = n - k - v_3(\xi) - 1 \geq m - k \geq (0.1075 + \eta/2)n$. The Renyi factor with the linear window costs $3^n \sum \mathbb{P}^2 \leq e^{\eta n}$ (Young's inequality converts the $C\sqrt{nj}$ window into an $e^{O(n)}$ factor absorbed by the slack, and Corollary 6.3-type injectivity survives since $\sum_j 3^{j-1} 2^{l-a[1,j]} < 3^n$ still closes). Assembling by Plancherel and Cauchy–Schwarz: $\mathrm{Osc} \leq \exp\big(-c(0.1075 + \frac{\eta}{2})n + \frac{\eta}{2} n \ln 2 + O(\log n)\big)$. The exponent $\varphi(\eta) = 0.1075c - \frac{\eta}{2}(\ln 2 - c)$ is decreasing in η (since $c < \ln 2$); the minimal feasible slack $\eta^* = 0.215c/(\ln 2 + c)$ (balancing the window constant against the slack) gives the stated $c_1 = \varphi(\eta^*)$. □

Remark 5.9 (The gate of Path B was an accounting artifact)**.** *There is no additive threshold on c: every positive Fourier exponent yields a positive mixing exponent. The price is multiplicative, $\kappa(c) = c_1/c = 0.215c/(\ln 2 + c) \in (0, 0.215)$, not a gate at 0.494.*

Corollary 5.10 (Conditional stabilization calculation)**.** *Assume $\mathrm{T0}_{\mathrm{cert}}$ with a genuine asymptotic exponent $c > 0$ and assume the remaining hypotheses in the fine-scale gluing argument. Then $d_{TV}(\mathrm{Pass}_x(\mathbf{N}_{x\alpha}), \mathrm{Pass}_x(\mathbf{N}_{x\alpha^2})) \leq Cx^{-\gamma_{\mathrm{Tao}}}$ with $\gamma_{\mathrm{Tao}} = c_1(c)(\alpha - 1)/100$. This is a conditional conversion formula only. The number 0.1438 is the ceiling of the white-point mechanism, not a certified Fourier exponent, and the finite-n value near 0.25 cannot be substituted as an asymptotic rate.*

5.1 Two Fourier regimes and the worst-case barrier

Observation 5.11 (Two Fourier decay regimes; numerical). *The Syracuse measure exhibits two numerically distinct Fourier behaviours:*

(i) ***Bulk (L^2).*** *The collision dimension satisfies $D_2 \to 1$ and the root-mean-square decay rate is $c_{\text{avg}} \approx 0.61$ (nats), reflecting chaotic mixing of the typical mass. The independent Front–C sample in Observation 5.3 likewise has black density close to 2ε and a substantially faster exponential-moment rate than the resonant sample.*

(ii) ***Worst case (L^∞).*** *The supremum $\sup_{3\nmid\xi} |\hat{\mu}_n(\xi)|$ is attained in the computations at a low-discrete-log frequency $\xi = 2^{s^*}$ with $s^*/n \to \log_2 3$. The worst-case rate c_∞ satisfies $c_\infty \le 0.054$ in the available fits; the data do not exclude $c_\infty = 0$ (subexponential decay). The relevant exceptional set is the truncated low-exponent family $\{2^s : 0 \le s \le Cn\}$, whose cardinality is $O(n)$ compared with $\varphi(3^n) = 2 \cdot 3^{n-1}$; the full orbit $\{2^s \bmod 3^n\}$ is the entire unit group and must not be called L^2-negligible.*

Remark 5.12 (Status of Observation 5.11). *Both items are numerical. The low-s identification lacks a theoretical explanation, the search at large n does not exhaust all unit frequencies, and c_∞ is not determined. Front–C alignment (Observation 5.4) is independent supporting evidence, but it does not turn either numerical maximizer into a theorem.*

Theorem 5.13 (Worst-case barrier; cf. Tao, Remark 1.18). *For ξ not divisible by 3, $|\mathbb{E}e^{-2\pi i\xi\,\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z})/3^n}| \le \mathrm{Osc}_{n-1,n}$. Hence any exponential fine-scale bound $\mathrm{Osc}_{m,n} \le Ce^{-cm}$ at $m = n-1$ is limited by the worst-case Fourier rate c_∞: if $c_\infty = 0$, Tao's superpolynomial bound cannot be upgraded to exponential.*

Remark 5.14 (L^2 optics cannot bypass the barrier). *Although the bulk L^2 decay is fast, the finest-scale oscillation $\mathrm{Osc}_{n-1,n}$ is bounded* below *by the worst-case Fourier coefficient (Theorem 5.13). Averaging over frequencies therefore cannot yield an exponential bound when the worst-case decay is subexponential; the barrier is intrinsic to the equivalence of fine-scale mixing and pointwise Fourier decay.*

5.2 Honest status of the programme

Theorem 5.15 (Conditional reverse-geometry consequence). *If the reverse-chord encoding were valid, the reverse-growth calculation would assign the candidate exponent*

$$\gamma_{\text{rev}} = \frac{I_{\text{rev}}(1.33)}{\log_2 3 - 1.33} \approx 0.993$$

to the forward exceptional set. However, as established in Remark 3.26, the encoding suffers an intrinsic entropy deficit. Therefore, this number is strictly a characteristic exponent of reverse growth geometry and cannot bound E_{N_0}.

Remark 5.16 (What remains open). *(i) The restart/transport hypothesis across scales is open and currently mechanism-free. Every candidate mechanism — pointwise Fourier decay, low-frequency/aggregate cancellation, spectral transversality — has been falsified by the unrestricted spectral diagnostics (Assumption 3.15, Lemma 3.19).*

(ii) The worst-case Fourier rate c_∞ is not determined. If $c_\infty = 0$, Tao's architecture yields a superpolynomial but not an exponential exceptional-set bound.

(iii) *The full Collatz conjecture* $\mathrm{Col}_{\min}(N) = 1\ \forall N$ *is a pointwise statement. It is not implied by any density bound and remains beyond the statistical methods developed here.*

6 Conclusion: Local Descent, Reverse Geometry, and the Open Bridges

The reverse thermodynamic calculation gives the dimensionless geometric ratio

$$\gamma_{\mathrm{rev}} = \frac{I_{\mathrm{rev}}(1.33)}{\log_2 3 - 1.33} \approx \frac{0.2532}{0.25496} \approx 0.993.$$

The closure of the Fourier route. The exact boundary-layer spectrum formula and the affine word structure are mathematically sound and robustly validated. However, the analytic path to Forward Shadowing via Fourier-based max bounds or L2 cancellation is conclusively closed. The unrestricted spectrum preserves macroscopic structural peaks, and the high-frequency “noise” retains $O(1)$ magnitude, explicitly failing the required $2^{-B/2}$ decay threshold. Consequently, neither pointwise Fourier suppression nor Spectral Transversality provides the global error cancellation necessary for an asymptotic proof. The numerical $2^{-B/2}$ decay observed in specific finite-scale bad sets is a local phenomenon reflecting specific topological properties of those sets, rather than a universal property of the endpoint measure. Any path forward must circumvent these Fourier obstructions entirely.

Computational frontier. The numerical substitutions at $N_0 = 2^{68}$ previously reported from $KN_0^{-0.993}$ are therefore conditional on the unproved reverse-chord encoding, not consequences of the reverse LDP alone. They should be read only as a scale illustration under that stronger hypothesis, not as a residual density established by the present work.

Honest status and limits of the proof. It is crucial to clarify the rigorous boundaries of our results:

1. **Rigorous 3-adic Thermodynamics:** Steps A and B prove the properties of the tilted 3-adic model strictly. The spectral gap, the Doob h-transformed LLN, and the variance of the conditioned process are rigorous theorems about the 3-adic topology, confirmed by numerical simulation up to $n = 7$.

2. **Two Open Bridges:** The forward route requires restart/transport of the endpoint distribution between local 2-adic LDP blocks. The reverse value $\gamma_{\mathrm{rev}} \approx 0.993$ additionally requires a combinatorial encoding of every forward exceptional integer by a reverse 4/3-shift chord. Neither bridge is proved.

3. **The Worst-Case Barrier:** The Fourier route (Tao’s architecture) unconditionally relies on the fine-scale mixing equivalence. If the worst-case Fourier decay is subexponential ($c_\infty = 0$), then the architecture yields only a superpolynomial exceptional-set bound, not an exponential one. The L^2 optics cannot bypass this barrier at the finest scale $m = n - 1$.

In conclusion, the project currently contains a rigorous finite-scale 2-adic counting bound, conditional thermodynamic models, and numerical Fourier and reverse-growth observations. The value $\gamma_{\mathrm{rev}} \approx 0.993$ is a reverse-geometric characteristic, not an established exceptional-set exponent. Establishing a forward bound requires restart/transport; attaching the reverse exponent requires reverse-chord encoding. The full Collatz conjecture remains beyond these density methods.